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# Time-o1: Time-Series Forecasting Needs Transformed Label Alignment

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## Abstract

Training time-series forecast models presents unique challenges in designing effective learning objectives. Existing methods predominantly utilize the temporal mean squared error, which faces two critical challenges: (1) label autocorrelation, which leads to bias from the label sequence likelihood; (2) excessive amount of tasks, which increases with the forecast horizon and complicates optimization. To address these challenges, we propose Time-o1, a transformation-augmented learning objective for training time-series forecasting models. The central idea is to transform the label sequence into decorrelated components with discriminated significance. Models are then trained to align the most significant components, thereby effectively mitigating label autocorrelation and reducing task amount. Extensive experiments demonstrate that Time-o1 achieves state-of-the-art performance and is compatible with various forecast models. Code is available at <https://github.com/Master-PLC/Time-o1>.

## 1 Introduction

Time-series forecasting involves predicting future data from historical observations [60, 23] and has been applied across diverse domains, such as air quality prediction in meteorology [28], user behavior analysis in e-commerce [3], and process monitoring in manufacturing [47, 50]. To build effective forecast models, there are two questions that warrant investigation: (1) *How to design a neural network architecture to encode historical observations*, and (2) *How to devise a learning objective to train the neural network*. Both are critical for model performance.

Recent research has primarily focused on developing neural network architectures [54, 59]. The key challenge lies in exploiting the autocorrelation in the historical sequences. To this end, various architectures have been proposed [24, 40, 51, 30], such as recurrent neural networks [10, 9, 53], convolutional neural networks [55, 52, 26], and graph neural networks [62, 2, 12]. The current progress is marked by a debate between Transformers and simple linear models. Transformers, equipped with self-attention mechanisms, offer superior scalability [27, 34, 29]. In contrast, linear models, which encapsulate temporal dynamics using linear layers, are straightforward to implement and often demonstrate strong performance [66, 42, 64, 61]. These advancements showcase the rapid evolution in neural architecture design for time-series forecasting.

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\*This work was done in the internship at Xiaohongshu Inc. Both authors have equal contribution.

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In contrast, the design of learning objectives has received less attention [49, 39, 17, 22]. Most existing methods employ the temporal mean squared error (TMSE) as the learning objective, which measures the step-wise discrepancy between forecast and label sequences [27, 34]. While being effective for various scenarios, it has two critical limitations. First, it is biased against the true likelihood of label sequence due to the presence of autocorrelation in the label sequence. Second, the number of prediction tasks increases with the forecast horizon, which complicates the optimization process since multitask learning is known to be challenging given excessive tasks [68, 25]. These limitations highlight the importance of developing more principled learning objectives for time-series forecasting.

To handle these challenges, we propose a transformation-augmented learning objective tailored for time-series forecasting. The key idea is to transform the label sequence into decorrelated components ranked by significance. By aligning the most significant decorrelated components, Time-o1 mitigates label autocorrelation and reduces the number of tasks.

Our main contributions are summarized as follows:

- We formulate two critical challenges in designing objectives for time-series forecasting: label autocorrelation that induces bias, and the excessive number of tasks that impedes optimization.
- We propose Time-o1, which transforms labels into decorrelated components with discriminated significance. By aligning the significant components, it addresses the two challenges above.
- We conduct comprehensive experiments to demonstrate Time-o1’s efficacy, consistently boosting the performance of state-of-the-art forecast models across diverse datasets.

## 2 Preliminaries

This paper focuses on the time-series forecasting problem [38]. By the way of preface, uppercase letters (e.g.,  $Y$ ) denote random variables, and bolded letters (e.g.,  $\mathbf{Y}$ ) denote matrices containing data or parameters. One key distinction warrants emphasis: we are concentrating on the design of learning objectives for training forecast models [49, 39], rather than on the design of neural network architectures to implement the forecast models [27, 66, 58].

Suppose  $X$  is a time-series dataset with  $D$  covariates, where  $X_n$  denotes the observation at the  $n$ -th step. At an arbitrary  $n$ -th step, the historical sequence is defined as  $L = [X_{n-H+1}, \dots, X_n] \in \mathbb{R}^{H \times D}$ , the label sequence is defined as  $Y = [X_{n+1}, \dots, X_{n+T}] \in \mathbb{R}^{T \times D}$ , where  $H$  is the historical length and  $T$  is the forecast horizon. The target of time-series forecasting is to train a model  $g : \mathbb{R}^{H \times D} \rightarrow \mathbb{R}^{T \times D}$  that generates accurate prediction sequence  $\hat{Y}$  approximating the label sequence.

There are two aspects to building forecast models: (1) neural network architectures that effectively encode historical sequences, and (2) learning objectives for training these neural networks. While this paper focuses on the learning objective, we provide a brief review of both aspects for contextualization.

### 2.1 Model architectures for time-series forecasting

Neural networks are widely employed for encoding historical sequences [31, 57] due to their ability to automatically model feature interactions and capture complex nonlinear autocorrelation effects [30, 13, 11]. Notable examples include recurrent neural networks (e.g., S4 [10], Mamba [9]), convolutional neural networks (e.g., SCINet [26], TimesNet [55], MICN [52]), and graph neural networks (e.g., MTGNN [33], StemGNN [2]), each tailored to encode the dynamics within input sequences. The current progress centers on the comparison between Transformer-based architectures and linear architectures. Transformers (e.g., PatchTST [34], iTransformer [27], FreeFormer [65]) exhibit substantial scalability with increasing data size but entail high computational costs. In contrast, linear architectures (e.g., DLinear [66], RLinear [42], OLinear [64], TimeBase [14]) are generally more efficient but less scalable with larger datasets and struggle to handle varying input lengths.

### 2.2 Learning objectives for time-series forecasting

Modern time-series models predominantly adopt the direct forecast paradigm, generating  $T$ -step forecasts simultaneously using a multi-output head [21, 27, 66]. The learning objective is typically

the temporal mean squared error (TMSE) between the forecast and label sequences, given by:

$$\mathcal{L}_{\text{tmse}} = \sum_{t=1}^T \left( Y_t - \hat{Y}_t \right)^2, \quad (1)$$

which is widely employed in recent studies (e.g., FreTS [63], iTransformer [27], FredFormer [36], DUET [40]). However, this objective has been shown to be biased due to the autocorrelation present in the label sequence [49]. To address this bias, one line of research advocates for shape alignment between the forecast and label sequences to exploit autocorrelation (e.g., Dilate [19] and Soft-DTW [4]). However, these methods lack rigorous theoretical guarantees for unbiased objective and empirical evidence of improved performance. Another notable approach involves computing the forecast error in the frequency domain [49, 22], which reduces bias with theoretical guarantees [49].

### 3 Methodology

#### 3.1 Motivation

The learning objective is a fundamental component in training effective forecast models, yet its importance remains underexplored. Existing approaches predominantly employ the TMSE in (1) as the objective [27, 34, 64]. This practice, however, encounters two fundamental limitations rooted in the characteristics of time-series forecasting task.

First, TMSE introduces bias due to autocorrelation. In time-series forecasting, observations exhibit strong dependencies on their past values [66], resulting in step-wise correlation in the label sequence. In contrast, TMSE treats the forecast of each step as an independent task, thereby neglecting these correlations. This mismatch makes TMSE biased with respect to the true likelihood of the label sequence, as presented in Theorem 3.1.

**Theorem 3.1** (Autocorrelation bias). *Given label sequence  $Y$  where  $\Sigma \in \mathbb{R}^{T \times T}$  denotes the step-wise correlation coefficient, the TMSE in (1) is biased compared to the negative log-likelihood of the label sequence, which is given by:*

$$\text{Bias} = \left\| Y - \hat{Y} \right\|_{\Sigma^{-1}}^2 - \left\| Y - \hat{Y} \right\|^2 - \frac{1}{2} \log |\Sigma|. \quad (2)$$

where  $\|v\|_{\Sigma^{-1}}^2 = v^\top \Sigma^{-1} v$ . The bias vanishes if different steps in  $Y$  are decorrelated.<sup>3</sup>

Second, TMSE poses optimization difficulties as the forecast horizon grows. The large forecast horizon is crucial for applications such as manufacturing (enabling comprehensive production planning [48, 46]) and transportation (enabling proactive traffic management [67, 32]). As TMSE treats each forecasted step as an independent task, a large horizon results in excessive tasks. However, optimization is known to be difficult given excessive tasks [68, 25], as gradients from different tasks often conflict [68, 25], impeding convergence and leading to suboptimal model performance.

Designing effective learning objectives to handle the two limitations is challenging. The previous work **FreDF** [49] proposes a frequency loss, which transforms the label and forecast sequences into frequency components and aligns them in the frequency domain. This approach is motivated by Theorem 3.1: bias vanishes if different components are decorrelated. However, the decorrelation of frequency components holds only when the forecast horizon  $T \rightarrow \infty$  (see Theorem 3.3 in [49]). In real-world settings with finite horizon, frequency components remain correlated, rendering FreDF ineffective in eliminating bias. Additionally, the optimization difficulty remains, since transforming to the frequency domain retains the label length. *Consequently, FreDF does not fully address the autocorrelation bias and the optimization difficulty.*

Given the critical role of objective in training forecast models and the limitations of existing methods, it is compelling to develop an innovative objective to address the limitations and advance forecast performance. Importantly, there are two questions that warrant investigation. *How to devise an objective that eliminates autocorrelation and reduces task amount? Does it improve forecast performance?*

<sup>3</sup>The pioneering work [49] identifies the bias under the first-order Markov assumption on the label sequence. This study generalizes this bias without the first-order Markov assumption.

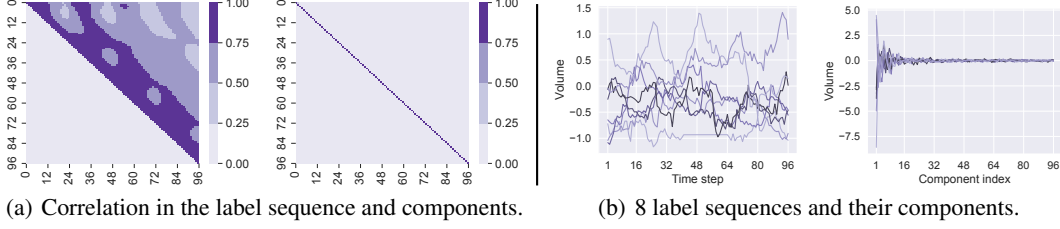


Figure 1: Comparison of label sequence and associated components. (a) shows the correlation volume within the label sequence (left panel) and components (right panel). (b) visualizes 8 label sequences randomly from ETTh1 (left panel) and the associated components (right panel).

### 3.2 Transforming label sequence with optimized projection matrix

In this section, we present a method for transforming label sequences into latent components to eliminate autocorrelation and distinguish significant components. Suppose  $\mathbf{Y} \in \mathbb{R}^{m \times T}$  contains normalized label sequences for  $m$  samples,  $\mathbf{P} = [\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_T]$  is the projection matrix, and the components are produced as  $\mathbf{Z} = \mathbf{Y}\mathbf{P}$ . The target is for  $\mathbf{Z}$  to be decorrelated and ranked by significance. For example, FreDF specifies  $\mathbf{P}$  as a Fourier matrix, which does not adapt to specific data properties and thus fails to decorrelate the components and distinguish the significant components<sup>4</sup>.

A natural approach to obtaining the projection matrix  $\mathbf{P}$  is solving optimization problem with constraints to ensure the desired properties. To find the  $p$ -th component, the projection vector can be calculated by solving the following problem:

$$\begin{aligned} \mathbf{P}_p^* = \underset{\mathbf{P}_p}{\operatorname{argmax}} \quad & (\mathbf{Y}\mathbf{P}_p)^\top (\mathbf{Y}\mathbf{P}_p) \\ \text{subject to} \quad & \begin{cases} \|\mathbf{P}_p\|^2 = 1 \\ \mathbf{P}_p^\top \mathbf{P}_j = 0, \forall j < p \quad \text{if } p > 1 \end{cases} \end{aligned} \quad (3)$$

where  $\mathbf{Z}_p = \mathbf{Y}\mathbf{P}_p$  is the  $p$ -th component, the normalization constraint  $\|\mathbf{P}_p\|^2 = 1$  is imposed to avoid trivial solution:  $\mathbf{P}_p \rightarrow \infty$ . The optimization target is to maximize the variance of  $\mathbf{Z}_p$ , which is equivalent to maximizing its significance, as components with larger variance contain richer information. For  $p > 1$ , the projection axis is required to be orthogonal to the previous axes to avoid redundancy. By solving the optimizations above from  $p = 1$  to  $T$  sequentially, we obtain the projection matrix  $\mathbf{P}^* = [\mathbf{P}_1^*, \dots, \mathbf{P}_T^*]$ . The components are then produced as  $\mathbf{Z} = \mathbf{Y}\mathbf{P}^*$ .

**Lemma 3.2** (Decorrelated components). *Suppose  $\mathbf{Y} \in \mathbb{R}^{m \times T}$  contains normalized label sequences for  $m$  samples,  $\mathbf{Z} = [\mathbf{Z}_1, \dots, \mathbf{Z}_T]$  are the obtained components; for any  $p \neq p'$ , we have  $\mathbf{Z}_p^\top \mathbf{Z}_{p'} = 0$ .*

**Lemma 3.3.** *The projection matrix  $\mathbf{P}^*$  can be obtained via singular value decomposition (SVD):  $\mathbf{Y} = \mathbf{U}\mathbf{\Lambda}(\mathbf{P}^*)^\top$ , where  $\mathbf{U} \in \mathbb{R}^{m \times m}$  and  $\mathbf{P}^* \in \mathbb{R}^{T \times T}$  consist of singular vectors, and the diagonal of  $\mathbf{\Lambda} \in \mathbb{R}^{m \times T}$  consists of singular values.*

**Theoretical Justification.** According to Theorem 3.1, the bias vanishes as the correlations between labels to be aligned are eliminated. The obtained components are decorrelated (Lemma 3.2), thereby mitigating autocorrelation-induced bias. Moreover, component significance decreases from  $\mathbf{Z}_1$  to  $\mathbf{Z}_T$  as they are derived by maximizing significance under sequentially augmented constraints. Furthermore,  $\mathbf{P}^*$  can be computed via SVD (Lemma 3.3), offering an efficient alternative to sequentially solving the constrained optimization problems in (3).

**Case study.** To showcase the implications of the obtained components, a case study was conducted on the ETTh1 dataset. Implementation details are provided in Appendix A. The results are illustrated in Fig. 1, with key observations summarized as follows:

- **Decorrelation effect:** Fig. 1 (a) compares the correlation volume in the label sequence and the generated components. In the left panel, the value at row  $i$  and column  $j$  represents the correlation

<sup>4</sup>In the subsequent paragraphs, we use the univariate case with  $D = 1$  for clarity. In the multivariate case, different variates can be treated separately to produce decorrelated components.

between the  $i$ -th and  $j$ -th steps. A large number of non-diagonal elements exhibit substantial values, with approximately 50.5% exceeding 0.25, indicating notable autocorrelation in the label sequence. In contrast, the right panel shows negligible values for the non-diagonal elements. This demonstrates that transforming the label sequence into components effectively eliminates correlation, thereby corroborating Theorem 3.2.

- **Significance discrimination:** Fig. 1 (b) compares the variance of the label sequence and associated components. In the left panel, the variance of different steps in the label sequence is relatively uniform, ranging from -1.5 to 1.5, suggesting that all steps are equally significant. In the right panel, however, only a few components exhibit large variance, while the others fluctuate within a narrow range. This indicates that the significance of different components can be clearly discerned, allowing for a trade-off between a slight loss of information and reduced optimization complexity by focusing on the most significant components.

The transformation is highly inspired by principal component analysis (PCA) [35]. However, one key distinction warrants emphasis. Existing works dominantly employ principal component analysis on *input features* for obtaining informative representations [8, 6], in contrast, we adapt it to *label sequence*, specifically aiming to reduce autocorrelation bias and simplify optimization for time-series forecasting. To our knowledge, this remains a technically innovative strategy.

### 3.3 Model implementation

In this section, we present the implementation details of Time-o1. The approach centers on extracting the latent components from the label sequence, then optimizing the forecast model using the most significant components.

Given an input historical sequence, the forecast model predicts a sequence  $\hat{\mathbf{Y}}$ . In line with prevailing preprocessing practices [27, 66, 36], label sequences are first standardized (step 1), which facilitates the decorrelation prerequisite specified in Lemma 3.2. Next, following Lemma 3.3, we compute the optimal projection by applying SVD to the label sequence. The matrix  $\mathbf{P}^*$ , composed of the right singular vectors, provides the required projections described in (3). Both forecasted and label sequences are then projected into the latent component space (step 3), where the first column carries the largest significance, successively diminishing across columns.

Suppose  $K$  is the number of retained components, the training objective using them is given by:

$$\mathcal{L}_{\text{trans},\gamma} := \left\| \hat{\mathbf{Z}}_{:,1:K} - \mathbf{Z}_{:,1:K} \right\|_1, \quad (4)$$

where  $K = \text{round}(\gamma \cdot T)$ , with  $\gamma$  controlling the involution ratio, the  $\ell_1$  norm  $\|\cdot\|_1$  computes the sum of element-wise absolute differences. Typically, we use the  $\ell_1$  norm instead of the squared norm following [49], considering that latent components typically vary greatly in scale (Fig. 1), which makes the squared norm unstable in practice. The  $\ell_1$  norm yields more stable and robust optimization.

Finally, the two objectives ( $\mathcal{L}_{\text{tmse}}$  and  $\mathcal{L}_{\text{trans},\gamma}$ ) are fused following [49], with  $0 \leq \alpha \leq 1$  controlling the relative contribution:

$$\mathcal{L}_{\alpha,\gamma} := \alpha \cdot \mathcal{L}_{\text{trans},\gamma} + (1 - \alpha) \cdot \mathcal{L}_{\text{tmse}}. \quad (5)$$

By projecting both forecasts and labels into decorrelated components, Time-o1 effectively reduces autocorrelation bias. By focusing exclusively on the most significant components, Time-o1 reduces optimization difficulty with minimal information loss. Time-o1 is model-agnostic, offering practitioners the flexibility to employ the most suitable forecast model for each specific scenario.

## 4 Experiments

To demonstrate the efficacy of Time-o1, there are six aspects empirically investigated:

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**Algorithm 1** The workflow of Time-o1.

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**Input:**  $\hat{\mathbf{Y}}$ : forecast sequences,  $\mathbf{Y}$ : label sequences.  
**Parameter:**  $\alpha$ : the relative weight of the transformed loss,  $\gamma$ : the ratio of involved significant components.  
**Output:**  $\mathcal{L}_{\alpha,\gamma}$ : the obtained learning objective.

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1:  $\mathbf{Y} \leftarrow \text{Standardize}(\mathbf{Y})$ .
2:  $\mathbf{P}^* \leftarrow \text{SVD}(\mathbf{Y})$ 
3:  $\mathbf{Z} \leftarrow \mathbf{Y}\mathbf{P}^*$ ,  $\hat{\mathbf{Z}} \leftarrow \hat{\mathbf{Y}}\mathbf{P}^*$ 
4:  $K \leftarrow \text{round}(\gamma \cdot T)$ 
5:  $\mathcal{L}_{\text{trans},\gamma} \leftarrow \|\hat{\mathbf{Z}}_{:,1:K} - \mathbf{Z}_{:,1:K}\|_1$ 
6:  $\mathcal{L}_{\text{tmp}} \leftarrow \|\hat{\mathbf{Y}} - \mathbf{Y}\|_2^2$ 
7:  $\mathcal{L}_{\alpha,\gamma} := \alpha \cdot \mathcal{L}_{\text{trans},\gamma} + (1 - \alpha) \cdot \mathcal{L}_{\text{tmse}}$ 

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Table 1: Long-term forecasting performance.

| Models  | Time-o1 (Ours) |              | Fredformer (2024) |              | iTransformer (2024) |              | FreTS (2023) |       | TimesNet (2023) |       | MICN (2023)  |              | TiDE (2023) |              | DLinear (2023) |       | FEDformer (2022) |       | Autoformer (2021) |       | Transformer (2017) |       |
|---------|----------------|--------------|-------------------|--------------|---------------------|--------------|--------------|-------|-----------------|-------|--------------|--------------|-------------|--------------|----------------|-------|------------------|-------|-------------------|-------|--------------------|-------|
|         | MSE            | MAE          | MSE               | MAE          | MSE                 | MAE          | MSE          | MAE   | MSE             | MAE   | MSE          | MAE          | MSE         | MAE          | MSE            | MAE   | MSE              | MAE   | MSE               | MAE   | MSE                | MAE   |
| ETTh1   | <b>0.380</b>   | <b>0.393</b> | <u>0.387</u>      | <u>0.398</u> | 0.411               | 0.414        | 0.414        | 0.421 | 0.438           | 0.430 | 0.396        | 0.421        | 0.413       | 0.407        | 0.403          | 0.407 | 0.442            | 0.457 | 0.526             | 0.491 | 0.799              | 0.648 |
| ETTh2   | <b>0.272</b>   | <b>0.317</b> | <u>0.280</u>      | <u>0.324</u> | 0.295               | 0.336        | 0.316        | 0.365 | 0.302           | 0.334 | 0.308        | 0.364        | 0.286       | 0.328        | 0.342          | 0.392 | 0.308            | 0.354 | 0.315             | 0.358 | 1.662              | 0.917 |
| ETTh1   | <b>0.431</b>   | <b>0.429</b> | <u>0.447</u>      | <u>0.434</u> | 0.452               | 0.448        | 0.489        | 0.474 | 0.472           | 0.463 | 0.533        | 0.519        | 0.448       | 0.435        | 0.456          | 0.453 | 0.447            | 0.470 | 0.477             | 0.483 | 0.983              | 0.774 |
| ETTh2   | <b>0.359</b>   | <b>0.388</b> | <u>0.377</u>      | 0.402        | 0.386               | 0.407        | 0.524        | 0.496 | 0.409           | 0.420 | 0.620        | 0.546        | 0.378       | <u>0.401</u> | 0.529          | 0.499 | 0.452            | 0.461 | 0.448             | 0.460 | 2.688              | 1.291 |
| ECL     | <b>0.170</b>   | <b>0.260</b> | 0.191             | 0.284        | <u>0.179</u>        | <u>0.270</u> | 0.199        | 0.288 | 0.212           | 0.306 | 0.192        | 0.302        | 0.215       | 0.292        | 0.212          | 0.301 | 0.214            | 0.328 | 0.249             | 0.354 | 0.265              | 0.358 |
| Traffic | <b>0.419</b>   | <b>0.280</b> | 0.486             | 0.336        | <u>0.426</u>        | <u>0.285</u> | 0.538        | 0.330 | 0.631           | 0.338 | 0.529        | 0.312        | 0.624       | 0.373        | 0.625          | 0.384 | 0.640            | 0.398 | 0.662             | 0.416 | 0.692              | 0.379 |
| Weather | <b>0.241</b>   | <b>0.280</b> | 0.261             | <u>0.282</u> | 0.269               | 0.289        | <u>0.249</u> | 0.293 | 0.271           | 0.295 | 0.264        | 0.321        | 0.272       | 0.291        | 0.265          | 0.317 | 0.326            | 0.372 | 0.319             | 0.365 | 0.699              | 0.601 |
| PEMS03  | <b>0.097</b>   | <b>0.208</b> | 0.146             | 0.260        | 0.122               | 0.233        | 0.149        | 0.261 | 0.126           | 0.230 | <u>0.106</u> | <u>0.223</u> | 0.316       | 0.370        | 0.216          | 0.322 | 0.152            | 0.275 | 0.411             | 0.475 | 0.122              | 0.226 |
| PEMS08  | <b>0.141</b>   | <b>0.237</b> | 0.171             | 0.271        | <u>0.149</u>        | <u>0.247</u> | 0.174        | 0.275 | 0.152           | 0.243 | 0.153        | 0.258        | 0.318       | 0.378        | 0.249          | 0.332 | 0.226            | 0.312 | 0.422             | 0.456 | 0.240              | 0.261 |

Note: We fix the input length as 96 following [27]. **Red** and underlined denote best and second-best results, respectively. Avg indicates average results over forecast horizons: T=96, 192, 336 and 720. Time-o1 employs the top-performing baseline on each dataset as its underlying forecast model.

1. **Performance:** *Does Time-o1 work?* We compare Time-o1 with state-of-the-art baselines using public datasets on long-term forecasting in Section 4.2 and short-term forecasting tasks in Appendix E.1. Moreover, we compare Time-o1 with other learning objectives in Section 4.3.
2. **Gain:** *How does it work?* Section 4.4 offers an ablative study to dissect the contributions of the individual factors of Time-o1, elucidating their roles in enhancing forecast accuracy.
3. **Generality:** *Does it support other forecast models?* Section 4.5 verifies the adaptability of Time-o1 across different forecast models, with additional results in Appendix E.4.
4. **Flexibility:** *Does it support alternative transformations?* Section 4.5 also investigates generating latent components with other transformations to showcase flexibility of implementation.
5. **Sensitivity:** *Does it require careful fine-tuning?* Section 4.6 presents a sensitivity analysis of the hyperparameter  $\alpha$ , where Time-o1 maintains efficacy across a broad range of parameter values.
6. **Efficiency:** *Is Time-o1 computationally expensive?* Section D investigates the running cost of Time-o1 in diverse settings.

## 4.1 Setup

**Datasets.** In this work, we conduct experiments on ETT (4 subsets), ECL, Traffic, Weather, and PEMS [27] for long-term forecasting task, and M4 for short-term forecasting task [55]. All datasets are split chronologically into training, validation, and testing sets following our prior work [49].

We compare Time-o1 with a range of established models, including Transformer [43], Autoformer [56], FEDformer [69], iTransformer [27], Fredformer [36], DLinear [66], TiDE [5], FreTS [63], TimesNet [55], and MICN [52]. As a learning objective, Time-o1 is model-agnostic and can be integrated with any model architecture. For fair comparison, Time-o1 employs the best-performing baseline model on each dataset as its underlying model architecture.

**Implementation.** The baseline models are reproduced using the scripts provided by Fredformer [36]. [37]. They are trained using the Adam [15] optimizer to minimize the TMSE loss. Datasets are split chronologically into training, validation, and test sets. Following the prestigious benchmark [37], the dropping-last trick is disabled during the test phase. When integrating Time-o1 to enhance an established model, we adhere to the associated hyperparameter settings in the public benchmark [36, 27], only tuning  $\alpha$ ,  $\gamma$  and learning rate conservatively. Experiments are conducted on Intel(R) Xeon(R) Platinum 8383C CPUs and NVIDIA RTX H100 GPUs.

## 4.2 Overall performance

Table 1 presents the long-term forecasting results. Time-o1 consistently improves base model performance. For example, on ETTh1, it reduces Fredformer’s MSE by 0.016. Similar gains



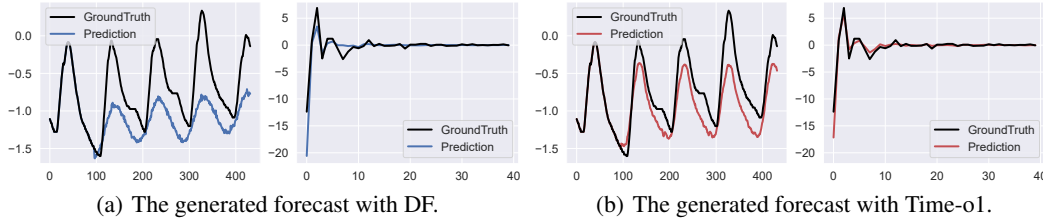


Figure 2: The visualization of forecast sequence generated by DF and Time-o1. The left panels in (a) and (b) present label and forecast sequences, the right panels present the associated components.

Table 2: Comparable results with other objectives for time-series forecast.

| Loss         |         | Time-o1      |              | FreDF        |              | Koopman |       | Dilate |       | Soft-DTW     |       | DPTA  |       | DF    |       |
|--------------|---------|--------------|--------------|--------------|--------------|---------|-------|--------|-------|--------------|-------|-------|-------|-------|-------|
| Metrics      |         | MSE          | MAE          | MSE          | MAE          | MSE     | MAE   | MSE    | MAE   | MSE          | MAE   | MSE   | MAE   | MSE   | MAE   |
| Fredformer   | ETTh1   | <b>0.379</b> | <b>0.393</b> | <u>0.384</u> | <u>0.394</u> | 0.389   | 0.400 | 0.389  | 0.400 | 0.397        | 0.402 | 0.396 | 0.402 | 0.387 | 0.398 |
|              | ETTTh1  | <b>0.431</b> | <b>0.429</b> | <u>0.438</u> | <u>0.434</u> | 0.452   | 0.443 | 0.453  | 0.442 | 0.460        | 0.449 | 0.460 | 0.449 | 0.447 | 0.434 |
|              | ECL     | <b>0.178</b> | <b>0.270</b> | <u>0.179</u> | <u>0.272</u> | 0.190   | 0.282 | 0.187  | 0.280 | 0.206        | 0.298 | 0.202 | 0.294 | 0.191 | 0.284 |
|              | Weather | <b>0.255</b> | <b>0.276</b> | <u>0.256</u> | <u>0.277</u> | 0.257   | 0.279 | 0.258  | 0.280 | 0.261        | 0.280 | 0.260 | 0.280 | 0.261 | 0.282 |
| iTransformer | ETTh1   | <b>0.395</b> | <b>0.401</b> | <u>0.405</u> | <u>0.405</u> | 0.413   | 0.416 | 0.407  | 0.412 | 0.417        | 0.415 | 0.416 | 0.415 | 0.411 | 0.414 |
|              | ETTTh1  | <b>0.438</b> | <b>0.434</b> | <u>0.442</u> | <u>0.437</u> | 0.455   | 0.451 | 0.452  | 0.448 | 0.470        | 0.457 | 0.463 | 0.454 | 0.452 | 0.448 |
|              | ECL     | <b>0.170</b> | <b>0.260</b> | 0.176        | <u>0.264</u> | 0.178   | 0.269 | 0.178  | 0.269 | <u>0.175</u> | 0.266 | 0.177 | 0.267 | 0.179 | 0.270 |
|              | Weather | <b>0.251</b> | <b>0.272</b> | <u>0.257</u> | <u>0.276</u> | 0.289   | 0.313 | 0.286  | 0.309 | 0.292        | 0.316 | 0.291 | 0.313 | 0.269 | 0.289 |

Note: **Bold** and underlined denote best and second-best results, respectively. The reported results are averaged over forecast horizons: T=96, 192, 336 and 720.

across other datasets further validate its effectiveness. These results suggest that modifying the learning objective can yield improvements comparable to, or even exceeding, those from architectural advancements. We attribute this to two key aspects of Time-o1: its decorrelation effect for debiased training, and its discrimination on significant components, which simplifies optimization.

**Showcases.** We visualize the forecast sequences and the generated components to showcase the improvements of Time-o1 in forecast quality. A snapshot on ETTm2 with historical window  $H = 96$  and forecast horizon  $T = 336$  is depicted in Fig. 2. Although the model trained using canonical DF captures general trends, its forecast struggles with large variations (e.g., peaks within steps 100-400). This reflects its difficulty in modeling significant, high-variance components. In contrast, Time-o1, by explicitly discriminating and aligning these significant components, generates a forecast that accurately captures these large variations, including the peaks within steps 100-400.

### 4.3 Learning objective comparison

Table 2 compares Time-o1 against other time-series learning objectives: FreDF [49], Koopman [18], Dilate [19], Soft-DTW [4], and DPTA [41]. For fair evaluation, we integrated their official implementations into both Fredformer and iTransformer.

Overall, shape alignment objectives (Dilate, Soft-DTW, DPTA) offer little performance gain over canonical DF (using TMSE loss), consistent with the findings in [19]. This phenomenon is rationalized by the fact that they do not mitigate the label autocorrelation nor reduce task amounts for simplifying optimization. FreDF improves performance by partly addressing autocorrelation bias. However, as discussed in Section 3.1, FreDF does not fully eliminate this bias, nor does it distinguish significant components to simplify the optimization landscape. Time-o1 directly addresses these two limitations of FreDF, leading to its superior overall performance.

### 4.4 Ablation studies

Table 3 presents an ablation study dissecting the contributions of critical factors in Time-o1: the decorrelation effect and the task reduction effect. The main findings are summarized as follows.

Table 3: Ablation study results.

| Model                | Decorrelation | Reduction | Data    | T=96         |              | T=192        |              | T=336        |              | T=720        |              | Avg          |              |
|----------------------|---------------|-----------|---------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                      |               |           |         | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          | MSE          | MAE          |
| DF                   | $\times$      | $\times$  | ETTh1   | 0.326        | 0.361        | 0.365        | 0.382        | 0.396        | 0.404        | 0.459        | 0.444        | 0.387        | 0.398        |
|                      |               |           | ETTh1   | 0.377        | 0.396        | 0.437        | 0.425        | 0.486        | 0.449        | 0.488        | 0.467        | 0.447        | 0.434        |
|                      |               |           | ECL     | 0.150        | 0.242        | 0.168        | 0.259        | 0.182        | 0.274        | 0.214        | 0.304        | 0.179        | 0.270        |
|                      |               |           | Weather | 0.174        | 0.228        | 0.213        | 0.266        | 0.270        | 0.316        | 0.337        | 0.362        | 0.249        | 0.293        |
| Time-o1 <sup>†</sup> | $\times$      | ✓         | ETTh1   | 0.338        | 0.366        | 0.369        | 0.383        | 0.397        | 0.403        | 0.458        | 0.441        | 0.391        | 0.398        |
|                      |               |           | ETTh1   | 0.376        | 0.395        | 0.437        | 0.430        | 0.478        | 0.450        | <u>0.469</u> | 0.467        | 0.440        | 0.436        |
|                      |               |           | ECL     | 0.150        | 0.239        | 0.164        | 0.253        | 0.178        | 0.268        | 0.210        | 0.296        | 0.175        | 0.264        |
|                      |               |           | Weather | <u>0.170</u> | <u>0.216</u> | 0.213        | 0.259        | 0.262        | <u>0.300</u> | 0.332        | <u>0.351</u> | 0.244        | <u>0.281</u> |
| Time-o1 <sup>‡</sup> | ✓             | $\times$  | ETTh1   | <u>0.324</u> | <u>0.359</u> | <u>0.362</u> | <u>0.379</u> | <u>0.390</u> | <u>0.400</u> | <u>0.451</u> | <u>0.438</u> | <u>0.382</u> | <u>0.394</u> |
|                      |               |           | ETTh1   | <u>0.373</u> | <u>0.395</u> | <u>0.433</u> | <u>0.423</u> | <u>0.476</u> | <u>0.445</u> | 0.474        | <u>0.463</u> | 0.439        | <u>0.431</u> |
|                      |               |           | ECL     | <u>0.147</u> | <u>0.238</u> | <u>0.162</u> | <u>0.252</u> | <u>0.174</u> | <u>0.267</u> | <u>0.205</u> | <u>0.294</u> | <u>0.172</u> | <u>0.263</u> |
|                      |               |           | Weather | 0.172        | 0.220        | <u>0.211</u> | <u>0.259</u> | <u>0.261</u> | 0.301        | <u>0.331</u> | 0.353        | <u>0.244</u> | 0.283        |
| Time-o1              | ✓             | ✓         | ETTh1   | <b>0.321</b> | <b>0.357</b> | <b>0.360</b> | <b>0.378</b> | <b>0.389</b> | <b>0.400</b> | <b>0.447</b> | <b>0.435</b> | <b>0.379</b> | <b>0.393</b> |
|                      |               |           | ETTh1   | <b>0.368</b> | <b>0.391</b> | <b>0.424</b> | <b>0.422</b> | <b>0.467</b> | <b>0.441</b> | <b>0.465</b> | <b>0.463</b> | <b>0.431</b> | <b>0.429</b> |
|                      |               |           | ECL     | <b>0.145</b> | <b>0.235</b> | <b>0.159</b> | <b>0.249</b> | <b>0.173</b> | <b>0.264</b> | <b>0.203</b> | <b>0.292</b> | <b>0.170</b> | <b>0.260</b> |
|                      |               |           | Weather | <b>0.169</b> | <b>0.219</b> | <b>0.210</b> | <b>0.258</b> | <b>0.259</b> | <b>0.297</b> | <b>0.327</b> | <b>0.349</b> | <b>0.241</b> | <b>0.280</b> |

Note: **Bold** and underlined denote best and second-best results, respectively.

Table 4: Varying transformations results.

| Transformation | ECL          |                |              |                | Weather      |                |              |                |
|----------------|--------------|----------------|--------------|----------------|--------------|----------------|--------------|----------------|
|                | MSE          | $\Delta$       | MAE          | $\Delta$       | MSE          | $\Delta$       | MAE          | $\Delta$       |
| None           | 0.179        | -              | 0.270        | -              | 0.249        | -              | 0.293        | -              |
| RPCA           | <u>0.171</u> | 4.31% ↓        | <u>0.261</u> | 3.16% ↓        | <u>0.244</u> | 1.78% ↓        | <u>0.286</u> | 2.38% ↓        |
| SVD            | 0.175        | 2.24% ↓        | 0.264        | 2.18% ↓        | 0.248        | 0.34% ↓        | 0.290        | 0.93% ↓        |
| FA             | 0.175        | 2.35% ↓        | 0.265        | 1.82% ↓        | 0.245        | 1.35% ↓        | 0.287        | 1.97% ↓        |
| Ours           | <b>0.170</b> | <b>4.86% ↓</b> | <b>0.260</b> | <b>3.57% ↓</b> | <b>0.241</b> | <b>2.94% ↓</b> | <b>0.280</b> | <b>4.28% ↓</b> |

Note:  $\Delta$  refers to the relative error reduction compared to the baseline (None). **Bold** and underlined denote best and second-best results.

- Time-o1<sup>†</sup> improves DF by reducing the number of tasks to optimize. To this end, it employs a randomized matrix as the projection matrix to generate components and aligns only a subset of the obtained components. The involution ratio  $\gamma$  is finetuned on the validation set. It consistently improves over DF (e.g.,  $-0.012$  MAE on Weather). This demonstrates that reducing tasks with a minimal loss of label information can reduce optimization difficulty and improve performance.
- Time-o1<sup>‡</sup> improves DF by aligning decorrelated components. To this end, the objective is calculated in (5) with  $\gamma = 1$ . It also outperforms DF, achieving the second-best results overall. This demonstrates aligning decorrelated label components to mitigate bias benefits forecast performance.
- Time-o1 integrates both factors above by aligning the most significant decorrelated components, which achieves the best performance, demonstrating the synergistic effect of these two factors.

#### 4.5 Generalization studies

In this section, we investigate the utility of Time-o1 with different transformation strategies and forecast models, to showcase the generality of Time-o1. In the bar-plots, the forecast errors are averaged over forecast lengths (96, 192, 336, 720), with error bars as 50% confidence intervals.

**Varying transformations.** We select alternative approaches to transform the label sequence into latent components and report the forecast performance in Table 4. The selected transformation methods include robust principal component analysis (RPCA) [1], SVD [7], and factor analysis [16]. Noting that the output of SVD yields components here, not a projection matrix as in Section 3.2. Implementation details are in Appendix C. Overall, all these transformation methods outperform canonical DF without transformation. However, the components obtained by these methods, including RPCA, cannot be guaranteed to be decorrelated. Consequently, autocorrelation bias may persist. In contrast, our approach ensures full decorrelation of the derived components (see Lemma 3.2), effectively addressing autocorrelation bias and leading to the best overall performance.

**Varying forecast models.** We explore the versatility of Time-o1 in augmenting representative forecast models: Fredformer [36], iTransformer [27], FreTS [63], and DLinear [66]. As illustrated in Fig. 3, Time-o1 improves forecast performance in all cases. For instance, on the Weather dataset, iTransformer and FreTS with Time-o1 achieve substantial reductions in MSE—up to 6.9% and 2.9%,



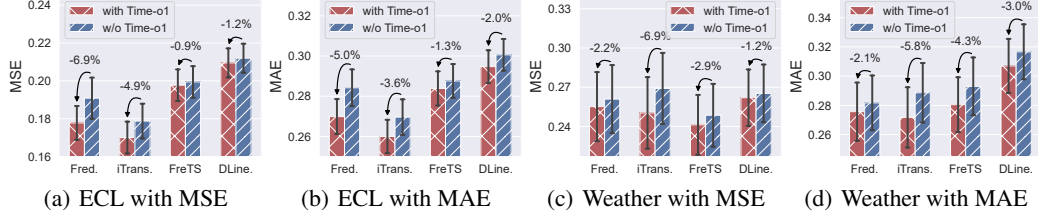


Figure 3: Improvement of Time-o1 applied to different forecast models, shown with colored bars for means over forecast lengths (96, 192, 336, 720) and error bars for 50% confidence intervals.

Table 5: Hyperparameter results on  $\alpha$ .

|          | ETTm1         |               | ETTh2         |               | Weather       |               |
|----------|---------------|---------------|---------------|---------------|---------------|---------------|
| $\alpha$ | MSE           | MAE           | MSE           | MAE           | MSE           | MAE           |
| 0        | 0.3867        | 0.3979        | 0.3766        | 0.4019        | 0.2486        | 0.2930        |
| 0.3      | 0.3871        | 0.3983        | 0.3742        | 0.3982        | 0.2439        | 0.2851        |
| 0.5      | 0.3864        | 0.3976        | 0.3703        | 0.3964        | <b>0.2432</b> | <b>0.2833</b> |
| 0.7      | <b>0.3831</b> | <u>0.3959</u> | <u>0.3674</u> | <b>0.3943</b> | <u>0.2433</u> | <u>0.2849</u> |
| 1        | <u>0.3850</u> | <b>0.3933</b> | <b>0.3606</b> | <u>0.3890</u> | 0.2753        | 0.3209        |

Note: **Bold** and underlined denote best and second-best results.

Table 6: Hyperparameter results on  $\gamma$ .

|          | ETTm1         |               | ETTh2         |               | Weather       |               |
|----------|---------------|---------------|---------------|---------------|---------------|---------------|
| $\gamma$ | MSE           | MAE           | MSE           | MAE           | MSE           | MAE           |
| 0.1      | 0.3915        | 0.4002        | 0.3816        | 0.4029        | <u>0.2437</u> | <u>0.2845</u> |
| 0.3      | 0.3849        | 0.3964        | 0.3694        | 0.3961        | <b>0.2424</b> | <b>0.2825</b> |
| 0.5      | 0.3817        | 0.3943        | 0.3651        | 0.3923        | 0.2466        | 0.2877        |
| 0.7      | <b>0.3798</b> | <b>0.3930</b> | <b>0.3603</b> | <b>0.3886</b> | 0.2443        | 0.2861        |
| 1        | <u>0.3814</u> | <u>0.3940</u> | <u>0.3624</u> | <u>0.3903</u> | 0.2491        | 0.2924        |

Note: **Bold** and underlined denote best and second-best results.

respectively. Further evidence of Time-o1’s versatility can be found in Appendix E.4. These results confirm Time-o1’s potential as a plug-and-play strategy to enhance adverse forecast models.

#### 4.6 Hyperparameter sensitivity

In this section, we examine the impact of critical hyperparameters on the performance of Time-o1. The results are presented in Table 5 and Table 6. Additional trends across different datasets and forecast lengths are provided in Appendix E.5. The primary observations are summarized as follows:

- The coefficient  $\alpha$  determines the relative importance of the transformed objective in (5). When  $\alpha$  is set to 1, Time-o1 exclusively uses the transformed objective. We observe that increasing  $\alpha$  from 0 to 1 generally leads to improved forecasting accuracy, with the best results typically achieved when  $\alpha$  is close to 1. The performance improvement is significant, e.g., MSE reduction on ETTh2 by 0.016, showcasing the utility of the transformed objective to improve forecast performance.
- The coefficient  $\gamma$  determines the ratio of involved components for alignment. When  $\gamma$  is set to 1, Time-o1 aligns all obtained components for model training. The results demonstrate that setting  $\gamma$  to 1, with all label information preserved, does not necessarily yield optimal performance. Instead, the best results are often obtained at  $\gamma < 1$ , rendering some loss of label information. For instance,  $\gamma = 0.7$  yields the best results on ETTm1 and ETTh2, while  $\gamma = 0.3$  is optimal for the Weather dataset. The rationale is that focusing on aligning the most significant components can reduce the task amount, thereby simplifying optimization. Since the majority of the information is contained in the most significant components, the information loss is minimal. Collectively, these factors contribute to improved forecast performance.

## 5 Conclusion

In this study, we highlight the importance of designing effective objectives for time-series forecasting. Two critical challenges are formulated: label autocorrelation, which induces bias, and the excessive number of tasks, which impedes optimization. To address these challenges, we introduce a model-agnostic learning objective called Time-o1. This method transforms the label sequence into decorrelated components with discernible significance. Forecast models are trained to align the most significant components, which effectively mitigates label autocorrelation due to the decorrelation between components and reduces task amount by discarding non-significant components. Experiments demonstrate that Time-o1 improves the performance of forecast models across diverse datasets.

**Limitations & future works.** In this work, we investigate the challenges of label autocorrelation and excessive number of tasks in time-series forecasting. Nevertheless, these issues also manifest in areas such as speech generation, target recognition, and dense image prediction. Applying Time-ol in these contexts is a promising avenue for future research. Additionally, historical sequence also exhibits autocorrelation and contains redundancy. Transforming inputs to derive decorrelated, compact representations could offer additional performance gains and also warrants investigation.

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